

Title: AI powered health focused robot

Student: Eric Choi

Faculty Sponsor: Zachary Butler

INTRODUCTION

This independent study focuses on the design and development of an AI-powered health-focused robot. The goal is to create a software system capable of performing basic health-related interactions through voice communication and sensor integration. These interactions may include context-aware health check-ins, symptom tracking, and escalation suggestions based on user input and vital readings.

The primary objective is to deepen my understanding of applied artificial intelligence, human-computer interaction, and systems-level software engineering. I will be collaborating with a mechatronics partner who will handle hardware design, while I lead all software development and integration efforts.

This project differs from standard course offerings by combining concepts from AI, HCI, embedded systems, and software architecture into a single cohesive application. It emphasizes cross-disciplinary collaboration, autonomous system design, and privacy-conscious interaction models—skills not covered comprehensively in any one existing course.

PLANNED WORK

The project consists of four major components: speech processing, rule-based AI logic, sensor data simulation and integration, and system orchestration. The work will be modular, with fallback simulation layers in case the hardware is delayed or limited.

Core CS topics to be explored:

- Rule-based decision systems and logical flow control
- Speech-to-text and text-to-speech pipelines
- Natural language processing and emotion-tagging logic
- Secure and privacy-conscious software architecture
- Integration of simulated and real-time sensor data

- Modular code design with interface abstraction and unit testing

Schedule:

Week 1–2: Finalize project scope and timeline, validate AI feature list, and confirm hardware direction with partner

Week 3–4: Set up development environment, test TTS/STT tools

Week 5–6: Implement rule-based health logic and simulate sensor inputs (e.g., temperature, heart rate)

Week 7: Midpoint check-in/demo using simulated inputs and logic

Week 8–9: Begin hardware integration (if components are ready), refine AI interaction logic

Week 10: Optional camera-based visual monitoring prototype

Week 11: Final integration and error handling

Week 12: Final documentation and testing

Week 13–14: Prepare and submit final project materials and presentation

- **Meeting Frequency:** Weekly 30-minute check-ins with faculty sponsor
 - **Estimated Time Commitment:** 5 hours per week, totaling ~65-70 hours over the semester
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DELIVERABLES

- A working AI system capable of voice interaction and simulated sensor-based health checks
 - Optional integration with real hardware for data input (if available)
 - A fully documented codebase with inline comments and a README
 - A final written report detailing design, implementation, testing, and key learnings
 - A live or recorded final demo
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EVALUATION

Evaluation will be based on:

- Quality and completeness of the final system (functionality, modularity, stability)
- Clarity and thoroughness of the documentation
- Consistency with the planned timeline and adaptability to challenges

- Final report and demonstration quality
- Active engagement and communication during weekly check-ins

Appendix to be included in the final report will outline specific tasks completed, time spent per task, and a total breakdown of project hours.